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Exam 4: NE533: Nuclear Fuel Performance
Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

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1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm.
   a) Estimate the oxide thickness after this time?
   b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume PBR = 1.56, ρ_{Zr} = 6.5 g/cc, ρ_{ZrO2} = 5.68 g/cc.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

1. a) $T = 625\ K$ Find t^*: $t^* = 6.62_{E}\text{-}7 \exp\left(\frac{11949\ K}{T}\right) d = 133.007\ d$

$t = 500\ d$ $t > t^* \rightarrow$ use linear relation

$t^0 = 600\ \mu m$ Find δ^*: $\delta^* = 5.1 \exp\left(\frac{-550\ k}{T}\right) \mu m = 2.11539\ \mu m$

Find K_L: $K_L = 7.48_{E}6 \exp\left(\frac{-12500\ k}{T}\right) \frac{\mu m}{d} = 0.0154174\ \frac{\mu m}{d}$

$$\delta = \delta^* + k_L(t - t^*) = 2.11539\ \mu m + 0.0154174\ \frac{\mu m}{d}(500\ d - 133.007\ d)$$

$$\Rightarrow \boxed{\delta = 7.7735\ \mu m}$$

b) $\delta = 7.7735\ \mu m$ $C_H^{tot} = C_H^0 + C_H^{clad} = 0 + C_H^{clad} \Rightarrow C_H^{tot} = C_H^{clad}$

$t^0 = 600\ \mu m$ $f_{ZrO_2}^{O} = \frac{A_{O_2}}{A_{Zr} + A_{O_2}} = \frac{32}{91+32} = 0.260163$

$PBR = 1.56$

$\rho_{ZrO_2} = 5.68\ \frac{g}{cm^3}$ $C_H^{tot} = C_H^{clad} = \frac{2f\delta\rho_{ZrO_2} f_{ZrO_2}^{O} \frac{M_H}{M_O}}{(t^0 - \frac{\delta}{PBR})\rho_{Zr}} \cdot 10^6$ wt. ppm

$\rho_{Zr} = 6.5\ \frac{g}{cm^3}$

$f = 0.14$ 18/18

$C_H^0 = 0$ wt. ppm

$$= \frac{2(0.14)(7.7735\ \mu m)(5.68\ \frac{g}{cm^3})(0.260163) \cdot \frac{1}{16}}{(600\ \mu m - \frac{7.7735\ \mu m}{1.56}) \cdot 6.5\ \frac{g}{cm^3}} \cdot 10^6 \text{ wt. ppm}$$

$$= \frac{0.201024\ \frac{\mu m\, g}{cm^3}}{3867.61\ \frac{\mu m\, g}{cm^3}} \cdot 10^6 \text{ wt. ppm}$$

$$= 51.9763 \text{ wt. ppm}$$

$$\Rightarrow \boxed{C_H^{tot} = 51.9763 \text{ wt. ppm}}$$

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2) Diffusion of oxygen through the oxide layer

8/8

3) Hydrides form preferentially around the rim (radially further out) of the cladding because hydrides prefer areas of high tensile stress and lower temperatures (the Soret effect). Hydrides are primarily embrittling; they decrease the cladding's ductility. They can also lower thermal conductivity.

4) A Reactivity Insertion Accident (RIA) is a type of accident that occurs when there is a large, unexpected increase (insertion) of reactivity.

In PWRs, this is typically a control rod ejection.

11/12

In BWRs, this is typically a control blade drop.

During a RIA, there is an immediate (~<0.1 s) increase in reactivity → fuel centerline T increases drastically → pressure in the rod increases drastically → failure occurs either by PCMI (the pellet physically fractures the clad) or DNB (temperatures too high to maintain cooling).

- melt

5) A Loss of Cooling Accident (LOCA) occurs when there is a loss of coolant flow. A LOCA is immediately mitigated with an emergency core cooling system, but still can cause:

12/12

- Rod inner pressure > Outer pressure → clad balloons and bursts
  → this ballooning further physically blocks coolant flow
- Oxidized Zr can shatter from thermal shock when exposed to ECCS water

It is different from a RIA in that it occurs on the order of several minutes (as opposed to milliseconds) and it has a completely different failure mode.

6) Higher burnup is generally more conducive to failure due to the gap being smaller, the oxide layer being thicker, there being a greater concentration of hydrides in the cladding, the pressure in the fuel rod being higher due to fission products / FGR, etc. Burnup increases failure probability. 4/4

- In a RIA, it makes PCMI failure more likely relative to in fresh fuel because the gap is smaller / potentially already closed.
- In a LOCA, it makes thermal shock embrittlement / shattering more likely because the cladding has a thicker oxide layer.

7) Steam oxidation resistance, fission product retention, improve fuels (reduce time to melt, temperatures), improve cladding (aqueous oxidation resistance, mechanical properties) 6/6

8) One near-term ATF concept is UO_2 doping. Through doping,
- UO_2 grain size can be increased, mitigating FGR as the gas takes longer to diffuse to the grain boundaries
- UO_2 thermal conductivity can be increased with the addition of certain metals 6/6

This mitigates two long-standing issues of UO_2.

9) Zr cladding, when exposed to high-temperature steam,
- loses most of its cooling ability and rapidly increases in temperature
- oxidizes at a much faster rate 5/6
- the oxide layer can crack away, revealing bare metal, severely increasing oxidation – called runaway or breakaway

~ most of it

10) Power to melt, cladding wear, cladding elongation and assembly bow, PCMI, cladding oxidation and hydrogen pickup (I listed 5 in case I misremembered some)

6/6

11) CRUD...
- decreases heat transfer (it's another layer between fuel + coolant)
- is a neutron poison (non-negligible neutron absorption cross-section)
- can flake off and cause out-of-core radiation sources (very bad for safety)

- absorbing boron

6/6

12) Zinc injection allows for Zn to outcompete ^{60}Co for tetrahedral sites and be deposited in the CRUD instead. Since Zn doesn't produce any harmful radiation, is less "flaky", and is less of a neutron poison, it is preferable to most metal species free in the water.

6/6

LiOH inclusion allows for a pH >6.9 to be maintained (which minimizes CRUD) even as boron (which decreases pH) is added to the fuel to control reactivity.

13) Formation of a Central Void A central void forms in MOX fuel pellets due to the higher temperatures making the center have preferential sites for vacancies

Restructuring From the rim moving inward, the grains go from as-sintered to equiaxed to columnar.

5/8

- this is two parts of the same phenomena